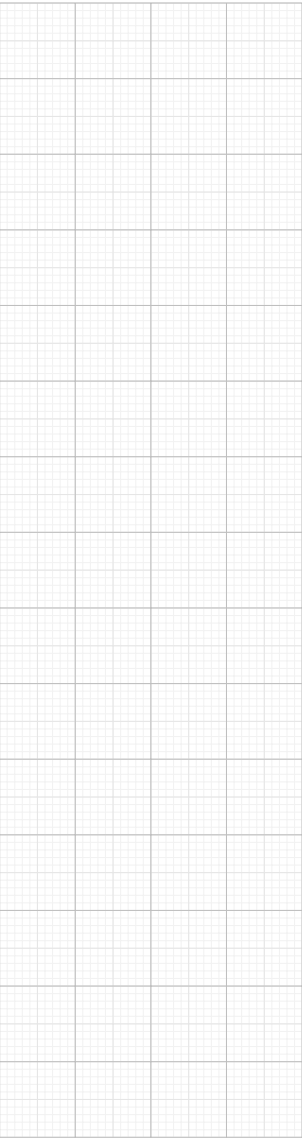


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Topics on Electrical Engineering

Drive Technology

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SS.2025
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B.Sc - Drive Technology

Magnetic Circuits and Materials

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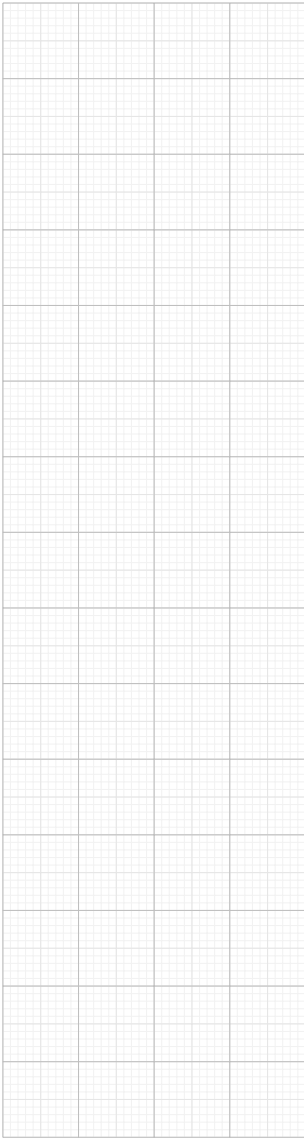
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Magnetic Circuits and Materials

- (LO1) An Overview of Maxwell's Equations,
- (LO2) Introduction to Magnetic Circuits,
- (LO3) Brief look on Magnetic Materials,
- (LO4) Magnetic Material Losses.



Learning Outcomes

Magnetic Circuits and Materials

- Maxwell's equations are a set of coupled partial differential equations which form the foundation of electric and magnetic circuits.
- In their PDE form, they are written as [1]:

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, & \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, & \nabla \times \mathbf{B} &= \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right).\end{aligned}$$

where $\mathbf{E}$ is the electric field, $\mathbf{B}$ is the magnetic field, ρ is the electric charge density and $\mathbf{J}$ the current density, ϵ_0 is the vacuum permittivity and μ_0 is the vacuum permeability.

Magnetic Circuits

Maxwell's Equations

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Magnetic Circuits and Materials

- Almost all electric drives use **ferromagnetic material** for shaping and directing $\mathbf{B}$ -fields.
 - $\mathbf{B}$ -fields act as the medium for **transferring/converting** energy.

Permanent Magnet (PM)s are also widely used in drive design.

- Without these materials, practical implementations of most familiar electro-mechanical devices would not be possible.
- Analysing and describing systems containing them is essential for designing effective drives.
- We start by looking at Maxwell's equations.

Introduction

Magnetic Materials

Magnetic Circuits and Materials

- Solution to Maxwell's equation can be hard if not attainable and usually simplifying assumptions are made to reach practical solutions.
- First, the displacement-current term ($\epsilon_0 \partial \mathbf{E} / \partial t$) can be neglected.

This term accounts for $\mathbf{B}$ -fields produced in space by time-varying $\mathbf{E}$ -fields (i.e., electromagnetic radiation) [2].

- Neglecting $\epsilon_0 \partial \mathbf{E} / \partial t$ results in the magneto-static form which relates $\mathbf{B}$ -fields to the currents ($\mathbf{J}$) which produce them.
- In integral form [3]: 6 / 56

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{a}, \quad \text{and} \quad \oint_S \mathbf{B} \cdot d\mathbf{a} = 0.$$

Magnetic Circuits

Maxwell's Equations

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These allow us to remove $\mathbf{E}$ field from our calculations.

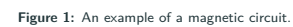
Maxwell's Equations

- This is similar to how currents are confined to the conductors of an electric circuit.

Maxwell's Equations

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Maxwell's Equations



Maxwell's Equations

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Magnetic Circuits and Materials

- The core is assumed to be composed of magnetic material with permeability much greater than of surrounding air (i.e., $\mu_r \gg \mu_0$).
- The core is of **uniform cross section** and is excited by a winding of N turns carrying a current of i amperes.

- This winding produces a **B -field** in the core.

Magnetic Circuits

Maxwell's Equations

Magnetic Circuits and Materials

- The source of the **B -field** in the core is the ampere-turn Ni .
- In magnetic circuit terminology Ni is the Magneto-motive Force (MMF) acting on the magnetic circuit.
- In systems with more than one winding, Ni must be replaced by the **algebraic sum** of the ampere-turns of all the windings:

$$\sum_{n=0}^{\infty} N_i i_i.$$

- The magnetic flux (ϕ) crossing a surface S is the surface integral of the normal component of **B** :

$$\phi = \int_S \mathbf{B} \cdot d\mathbf{a}$$

where the unit of flux (ϕ) is Wb.

Magnetic Circuits

Maxwell's Equations

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Magnetic Circuits and Materials

- Due to high permeability of the material (i.e., $\mu_r \gg \mu_0$), the magnetic flux is confined **almost entirely to the core**,
- The **B -field** lines follow the path defined by the core,
- The flux density is essentially uniform over a cross section as the cross-sectional area is **uniform**.
- The **B -field** can be visualised in terms of flux lines which form closed loops interlinked with the winding.

Magnetic Circuits

Maxwell's Equations

Magnetic Circuits and Materials

$$\phi = \int_S \mathbf{B} \cdot d\mathbf{a}$$

- This equation states that all flux entering the surface enclosing a volume must leave the volume as they form closed loops.
- This is enough to justify the uniformity of the magnetic flux density (**B**) across the cross section of a magnetic circuit (A_c):

$$\phi_c = B_c A_c,$$

where ϕ_c is the flux, B_c is the magnetic flux density magnitude and A_c is the cross-sectional core area.

Magnetic Circuits

Maxwell's Equations

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Magnetic Circuits and Materials

- Using the original magneto-static simplifications (i.e., no E -fields), we can come up to:

$$\mathcal{F} = Ni = \oint \mathbf{H} d\mathbf{l}.$$

- As the path of the flux line is close to the mean length¹ of the core (l_c), this is simplified to:

$$\mathcal{F} = Ni = H_c l_c,$$

where H_c is the average magnitude of $\mathbf{H}$.

Magnetic Circuits

Maxwell's Equations

Magnetic Circuits and Materials

- Transformers are wound on closed cores.
- However, electro-mechanical devices which incorporate a moving element must have air gaps in their magnetic circuits.
- When the air-gap (g) is much smaller than the dimensions of the adjacent core faces, the magnetic flux will follow the path defined by the core and the air gap and the techniques of magnetic-circuit analysis can be used.

If g becomes excessively large, the flux will be leak out of the sides of the air gap and the techniques of magnetic-circuit analysis will no longer be strictly applicable.

Magnetic Circuits

Maxwell's Equations

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Magnetic Circuits and Materials

- The relationship between the $\mathbf{H}$ and $\mathbf{B}$ is a property of the material.
- It is common to assume a linear relationship shown as:

$$\mathbf{B} = \mu \mathbf{H},$$

where μ is the magnetic permeability.

- In SI units, $\mathbf{H}$ is measured in $A \cdot m^{-1}$ and $\mathbf{B}$ is in $Wb \cdot m^{-2}$ or T.

Magnetic Circuits

Maxwell's Equations

Magnetic Circuits and Materials

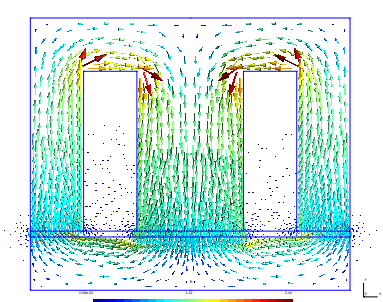


Figure 2: A Magnetic Circuits with two mmf sources.

Magnetic Circuits

Maxwell's Equations

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Magnetic Circuits and Materials

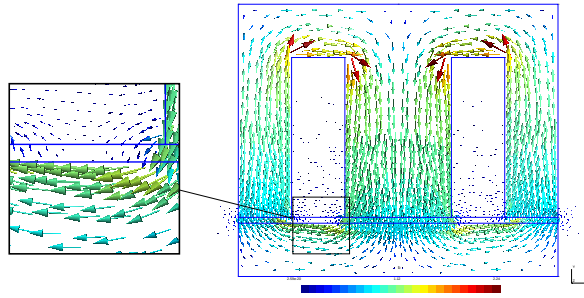


Figure 3: A closeup of the fringe paths caused by the air gap.

Magnetic Circuits Maxwell's Equations

Magnetic Circuits and Materials

- Applying this to the magnetic circuit gives:

$$\mathcal{F} = H_c l_c + H_g g,$$
 using the **B-H** relationship:

$$\mathcal{F} = \frac{B_c}{\mu} l_c + \frac{B_g}{\mu_0} g.$$
- Here, $\mathcal{F}$ is the MMF applied to the magnetic circuit where $\mathcal{F}_c = H_c l_c$ is the magnetic field in the core and $\mathcal{F}_g = H_g l_g$ is the magnetic field in the air-gap.
- Using ϕ we can re-write this equation as:

$$\mathcal{F} = \phi \left(\frac{l_c}{\mu A_c} + \frac{g}{\mu_0 A_g} \right).$$

Magnetic Circuits Maxwell's Equations

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Magnetic Circuits and Materials

- Provided the air-gap length g is sufficiently small, the configuration can be analysed as a magnetic circuit with two (2) series components:
 - Magnetic core (μ), cross-sectional area A_c , and mean length l_c ,
 - Air gap (μ_0), cross-sectional area A_g , and length l_g .
- In the core the flux density can be assumed uniform ($\mu \gg \mu_0$):

$$B_c = \frac{\phi}{A_c}, \quad \text{and in the air gap, } B_g = \frac{\phi}{A_g},$$

Magnetic Circuits Maxwell's Equations

Magnetic Circuits and Materials

- The term which multiplies mmf to flux is called **Reluctance**.

$$\mathcal{F} = \phi \left(\underbrace{\frac{l_c}{\mu A_c}}_{\mathcal{R}_c} + \underbrace{\frac{g}{\mu_0 A_g}}_{\mathcal{R}_g} \right) \quad \text{or} \quad \mathcal{F} = \phi (\mathcal{R}_c + \mathcal{R}_g)$$

Reluctance can be seen as analogous to resistance to electrical circuits.

Magnetic Circuits Maxwell's Equations

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Magnetic Circuits and Materials

- When a $\mathbf{B}$ -field varies with time, an $\mathbf{E}$ -field is produced in space as determined by Faraday's law [1]:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\oint_C \mathbf{E} \cdot d\mathbf{s} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}.$$

The line integral of the electric field intensity $\mathbf{E}$ around a closed contour C is equal to the time rate of change of the magnetic flux linking (i.e. passing through) that contour.

In electric drives, the effects of $\mathbf{E}$ field can be neglected.

Flux Linkage, Inductance and Energy

Magnetic Circuits and Materials

- As winding links the core flux N times, our equation is simplified to:

$$\mathcal{E} = N \frac{d\phi}{dt} = \frac{d\lambda}{dt} \quad \text{where} \quad \lambda = N\phi \quad \blacksquare$$

where λ is flux linkage, measured in (Wb · Turn),

Flux Linkage, Inductance and Energy

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Magnetic Circuits and Materials

- This simplification allows us to cancel the LHS of the equation to just the induced voltage (i.e., electromotive force)

$$\oint_C \mathbf{E} \cdot d\mathbf{s} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}.$$

- Whereas on the RHS, the flux is dominated by the core flux (ϕ).

$$\mathcal{E} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}.$$

Flux Linkage, Inductance and Energy

Magnetic Circuits and Materials

- For a magnetic circuit composed of magnetic material of constant magnetic permeability, the relationship between ϕ and i will be **linear** and we can define the inductance L as:

$$L = \frac{\lambda}{i} \quad \text{which can be written as} \quad L = \frac{N^2}{\mathcal{R}_{\text{tot}}}$$

where L is the inductance measured in Henry (H).

Inductance requires a linear relationship between flux and mmf.

Flux Linkage, Inductance and Energy

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Magnetic Circuits and Materials

Figure 4: An example of a magnetic winding circuit with 2 windings.

Flux Linkage, Inductance and Energy

Magnetic Circuits and Materials

- This resultant ϕ determines the operating point of the core material.
- If ϕ is broken up into terms attributable to the individual currents, the resultant flux linkages of coil 1 can be expressed as:

$$\lambda_1 = N_1 \phi = \overbrace{N_1^2 \left(\frac{\mu_0 A_c}{g} \right)}^{L_{11}} i_1 + \overbrace{N_1 N_2 \left(\frac{\mu_0 A_c}{g} \right)}^{L_{12}} i_2,$$

and this is simplified to:

$$\lambda_1 = L_{11} i_1 + L_{12} i_2,$$

where L_{11} is the **self** inductance and L_{12} is the **mutual** inductance of the coil.

Flux Linkage, Inductance and Energy

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Magnetic Circuits and Materials

- The MMF ($\mathcal{F}$) is given by the total ampere-turns.
- The reference directions for the currents (i_1, i_2) have been chosen to produce flux in the same direction.
- The total mmf is therefore:

$$\mathcal{F} = N_1 i_1 + N_2 i_2.$$

- With the reluctance of the core neglected and assuming that $A_c = A_g$, the core flux ϕ is:

$$\phi = (N_1 i_1 + N_2 i_2) \frac{\mu_0 A_c}{g},$$

where ϕ is the **resultant core flux** produced by the total mmf of the two windings.

Flux Linkage, Inductance and Energy

Magnetic Circuits and Materials

- Similarly, the flux linkage on coil 2 is:

$$\lambda_2 = N_2 \phi = \overbrace{N_1 N_2 \left(\frac{\mu_0 A_c}{g} \right)}^{L_{21}} i_1 + \overbrace{N_2^2 \left(\frac{\mu_0 A_c}{g} \right)}^{L_{22}} i_2$$

and this is simplified to:

$$\lambda_2 = L_{21} i_1 + L_{22} i_2$$

where $L_{21} = L_{12}$ is mutual inductance and L_{22} is self inductance.

It is important to note the ϕ is calculated based on the superposition of i_1, i_2 which implies a linear relationship between flux and mmf.

Flux Linkage, Inductance and Energy

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Magnetic Circuits and Materials

- It cannot be rigorously applied in situations where the **nonlinear characteristics of magnetic materials** dominate.
- However, in many situations of practical interest, the reluctance of the system is dominated by that of an air gap (which is of course linear) and the nonlinear effects of the magnetic material can be ignored.
- In other cases it may be perfectly acceptable to assume an average value of magnetic permeability for the core material.
- Using this it is possible to calculate a corresponding average inductance which can be used for calculations of reasonable engineering accuracy.

Flux Linkage, Inductance and Energy

Magnetic Circuits and Materials

- Magnetic materials can be used to **constrain and direct B -fields** in well-defined paths.
- In a transformer, they are used to maximize the coupling between the windings as well as to **lower the excitation current** required for transformer operation.
- In electric drives, they are used to shape the fields to obtain desired torque-production and electrical terminal characteristics.

Magnetic MaterialsIntroduction

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Magnetic Circuits and Materials

- Electro-mechanical devices require two (2) important properties:
 - Obtain large magnetic flux densities ($\uparrow B$)
 - Requiring relatively low levels of magnetizing force ($\downarrow H$).

Since magnetic forces and energy density increase with increasing flux density, this effect plays a large role in the performance of energy-conversion devices.

Magnetic MaterialsIntroduction

Magnetic Circuits and Materials

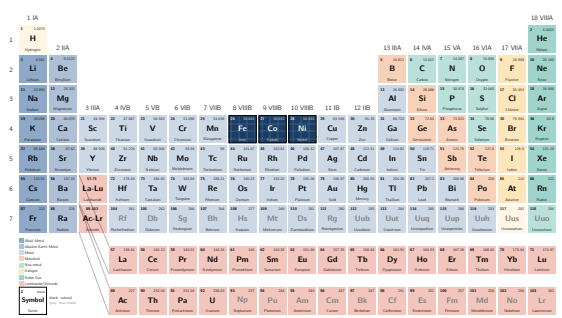


Figure 5: The periodic table of elements. Of all the elements that exist only a few of them have the ability to generate a magnetic field without any external need. The elements which exhibit ferromagnetism has been highlighted.

Magnetic MaterialsFerromagnetism

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Magnetic Circuits and Materials

- Typically composed of iron (**Fe**) and alloys of iron with:
 - Cobalt, Tungsten, Nickel, Aluminium, and other metals
- These materials are characterised by a wide range of properties,
 - Phenomena responsible for their properties **are the same**.
- Ferro-magnetic materials are found to be composed of a large number of **domains**.
 - Regions where the magnetic moments of all the atoms are parallel, giving rise to a net magnetic moment for that domain.
- In an un-magnetised sample of material, the domain magnetic moments are **randomly** oriented, and the net resulting magnetic flux in the material is zero (**0**).

Magnetic Materials

Ferromagnetism

Magnetic Circuits and Materials

- When **B**-field is applied to the material, the magnetic domain moments tend to **align** with the applied **B**-field.
- As a result, the magnetic moments add to the **B**-field, producing a higher flux density than would exist due to the magnetising force alone.
- Therefore the effective permeability μ , equal to the ratio of the total magnetic flux density to the applied magnetic-field intensity, is large compared with the permeability of free space (μ_0).
- As the magnetising force (**H**) is increased, this continues until all the magnetic moments are aligned with the applied field.
 - at this point they can no longer contribute to increasing the magnetic flux density

Magnetic Materials

Ferromagnetism

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Material becomes fully saturated.

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Magnetic Circuits and Materials

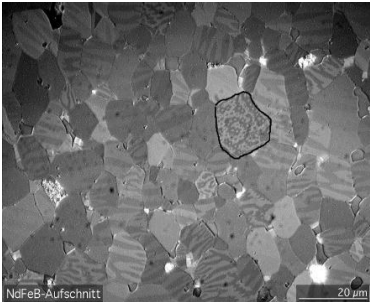


Figure 6: Microcrystalline grains within a piece of Nd2Fe14B (the alloy used in neodymium magnets) with magnetic domains made visible with a Kerr microscope. The domains are the light and dark stripes visible within each grain. The outlined grain has its magnetocrystalline axis almost vertical, so the domains are seen end-on [5].

Magnetic Materials

Ferromagnetism

Magnetic Circuits and Materials

- Without an external **B**-field, magnetic domains naturally align along certain directions associated with the **crystal structure** of the domain.
 - This is known as axes of easy magnetisation [6].
- If the applied **B** is reduced, the domain magnetic moments relax to the direction of easy magnetism nearest to the applied field.
- As a result, **B** is reduced to zero (**0**), although they will tend to relax towards their initial orientation, the magnetic dipole moments will no longer be totally random in their orientation;
- They will retain a net magnetisation **along the applied field**.
- It is this effect which is responsible for the phenomenon known as magnetic hysteresis [7].

Magnetic Materials

Ferromagnetism

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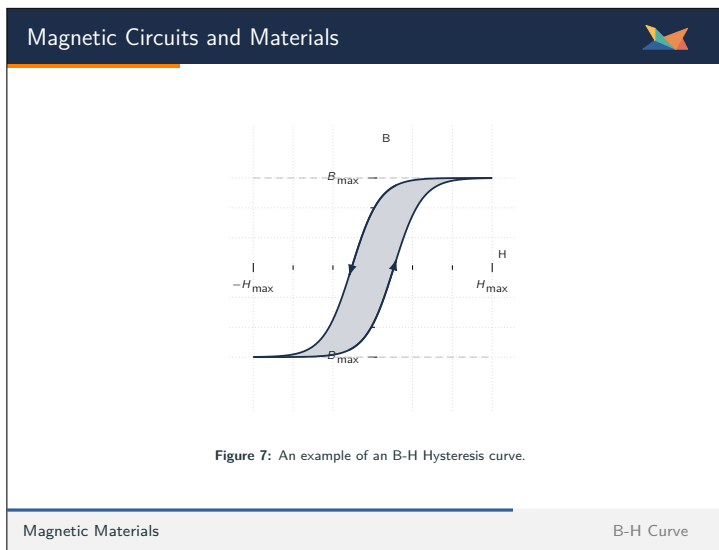
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Drive Technology

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Magnetic Circuits and Materials

1st Quadrant - Saturation

- Eventually the **B**-field will become high enough where no more change in the magnetisation occurs.
 - This is called **technical saturation**.
- It is possible to reach 99+% saturation.
- As the field is backed off from "saturation", the magnetisation declines very slightly to the B_r point.
 - This is the remanence, or residual induction [10].

All of the magnetic energy is now in the magnet and its field.

Magnetic Materials B-H Curve

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Magnetic Circuits and Materials

1st Quadrant

- The domains are very small, below the single domain size where there is resistance to demagnetization [8].
- The domains start to increase in size.

For a small interval, the magnetization will be reversible.

- As the field increases, magnetization will no longer reverse to zero but move on a minor hysteresis loop.
- Eventually the curve starts to bend over to the right.
 - It will still increase as more magnetic domains reach their full size and their magnetisation become parallel to the external field [9].

Magnetic Materials B-H Curve

Magnetic Circuits and Materials

2nd Quadrant - Demagnetisation

- In this quadrant the applied field **opposes** the materials **B**-field.
- As the external **B**-field increases in magnitude, some domains will reverse.
- At the knee of the demagnetisation curve, this increase has become rapid and the magnetisation will fall to the H_{ci} point.
- At H_{ci} , the number of domains aligned with the original magnetization is the same as the number aligned with the opposing magnetic field.

The net magnetisation is zero (0).

Magnetic Materials B-H Curve

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Magnetic Circuits and Materials

3rd Quadrant - Re-magnetisation

- The total magnetization of the part will be **reversed**.
- If we go far enough, magnetization will reach the saturation level in the negative direction.

4th Quadrant - Demagnetisation

- After fully reversing the magnet and removing the field in the third quadrant, magnetization will recoil to a point that is the negative of the B_r observed when in the first and second quadrant.
- If we apply additional field in the positive direction, we duplicate the second quadrant curve.

Magnetic Materials

B-H Curve

Magnetic Circuits and Materials

- From, the voltage induced in the N -turn winding is:

$$e(t) = \omega N \phi_{\max} \cos(\omega t) = E_{\max} \cos(\omega t)$$

where:

$$E_{\max} = \omega N \phi_{\max} = 2\pi f N A_c B_{\max}$$

- In steady-state AC, it is more important to use rms rather than instant values.
- Generally, the rms value of a periodic function of time $f(t)$, of T is:

$$F_{\text{rms}} = \sqrt{\left(\frac{1}{T} \int_0^T f^2(t) dt \right)}$$

AC Excitation

Definitions

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Magnetic Circuits and Materials

- In AC power systems, voltage and flux wave-forms closely approximate sinusoidal functions over time.
- Time to describe the excitation characteristics and losses associated with steady-state ac operation of magnetic materials.
- Assume a closed-core magnetic circuit (i.e., with no air gap),
- And a sinusoidal variation of the core flux (ϕ) with the following:

$$\phi(t) = \phi_{\max} \sin(\omega t) = A_c B_{\max} \sin(\omega t)$$

where:

 $\phi_{\max}$ amplitude of core flux in Wb, $B_{\max}$ Amplitude of flux density B_c in T, ω Angular frequency in rad/s, f Frequency in Hz.

AC Excitation

Definitions

Magnetic Circuits and Materials

- The rms value of a sine wave can be shown to be $1/\sqrt{2}$ times its peak value. Therefore the induced voltage rms is:

$$E_{\text{rms}} = \frac{2\pi}{\sqrt{2}} f N A_c B_{\max} = \sqrt{2} \pi N A_c B_{\max} = 4.44 N A_c B_{\max} \quad \blacksquare$$

- To produce magnetic flux in the core requires current in the exciting winding known as the exciting current, i_ϕ ,

The nonlinear magnetic properties of the core require that the wave-form of the exciting current differs from the sinusoidal waveform of the flux.

- When the hysteresis curve saturates the excitation current spikes.

AC Excitation

Definitions

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Magnetic Circuits and Materials

Figure 8: An Example Visualisation of the excitation current experienced by a magnetic circuit.

AC ExcitationDefinitions

Magnetic Circuits and Materials

- There are two (2) mechanisms associated with time varying **B**-fields.

Ohmic Heating

- Associated with induced currents in the core material.
- From Faraday's law, we see that **B**-fields give rise to **E**-fields.
- In magnetic materials these **E**-fields result in induced currents,
 - These are known as **eddy currents**, circulating in the core material, opposing the material's internal **B**-field [11].

AC ExcitationDefinitions

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Magnetic Circuits and Materials

- The exciting current supplies the mmf required to produce the core flux and the power input associated with the energy in the magnetic field in the core.
- Part of this energy is dissipated as losses and results in heating of the core.
- The rest appears as reactive power associated with energy storage in the magnetic field.

This reactive power is not dissipated in the core as it is cyclically supplied and absorbed by the excitation source.

AC ExcitationDefinitions

Magnetic Circuits and Materials

- To counteract the corresponding demagnetizing effect by the **eddy currents**, the current in the exciting winding must increase.

Therefore the resultant "dynamic" B-H loop under ac operation is somewhat "fatter" than the hysteresis loop for slowly varying conditions, and this effect increases as the excitation frequency is increased.

- It is for this reason that the characteristics of electrical steels vary with frequency and hence manufacturers typically supply characteristics over the expected operating frequency range of a particular electrical steel.

AC ExcitationDefinitions

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Magnetic Circuits and Materials

- To reduce the effects of eddy currents, magnetic structures are usually built of **thin sheets of laminations** of the magnetic material.
- These laminations, which are aligned in the direction of the field lines, are insulated from each other by an oxide layer on their surfaces or by a thin coat of insulating enamel or varnish.
- This greatly reduces the magnitude of the eddy currents since the layers of insulation interrupt the current paths; the thinner the laminations, the lower the losses.

In general, eddy-current loss tends to increase as the square of the excitation frequency and also as the square of the peak flux density.

AC Excitation

Definitions

Magnetic Circuits and Materials

- This is the required energy to move around the magnetic dipoles in the material and is dissipated as heat in the material.
- Therefore for a given flux level, the corresponding hysteresis losses are **proportional to the area of the hysteresis loop** and to the **total volume of material**.
- As there is an energy loss per cycle, hysteresis power loss is **proportional to the frequency of the applied excitation**. (Data Sheet)

AC Excitation

Hysteresis Losses

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Magnetic Circuits and Materials

- Due to the **hysteretic nature** of magnetic material.
- In a magnetic circuit or the transformer, a time-varying excitation (i_ϕ) will cause the magnetic material to undergo a cyclic variation described by a hysteresis loop.
- The energy input W to the magnetic core as the material undergoes a single cycle is shown to be:
$$W = \oint i_\phi d\lambda = \oint \left(\frac{H_c l_c}{N} \right) (A_c N dB_c) = A_c l_c \oint H_c dB_c$$
- Notice $A_c l_c$ is the **core volume** and the integral is the area of the ac hysteresis loop, we see that each time the magnetic material undergoes a cycle, there is a net energy input into the material.

AC Excitation

Hysteresis Losses

Magnetic Circuits and Materials

- In general, both losses depend on the metallurgy of the material as well as the flux density and frequency.
- Information on core loss is typically presented in graphical form.
- It is plotted in terms of watts per unit weight as a function of flux density;
 - Often a family of curves for different frequencies are given.
 - Generally it is either 50 or 60 Hz.

AC Excitation

Hysteresis Losses

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Magnetic Circuits and Materials

- Nearly all transformers and certain sections of electric machines use sheet-steel material that has highly favorable directions of magnetization along which the core loss is low and the permeability is high.
- This material is termed **grain-oriented steel** [12].
- The reason lies in the atomic structure of a crystal of the silicon-iron alloy,
 - which is a body-centred cube.

AC Excitation

Hysteresis Losses

Transformers

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Magnetic Circuits and Materials

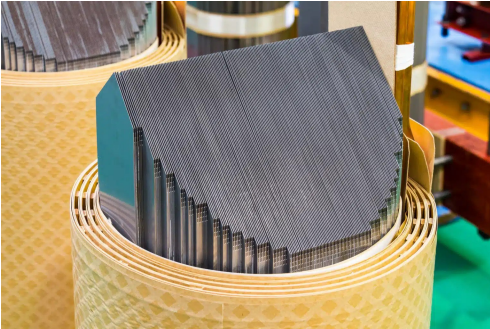


Figure 9: An example of Grain Oriented Steel used in industry [13].

AC Excitation

Hysteresis Losses

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Transformers

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Transformers

Learning Outcomes

(LO1) Modelling an ideal transformer,
(LO2) Construction methods,
(LO3) Physical properties,
(LO4) Types of transformers.



Introduction

Learning Outcomes

Transformers



Figure 10: An example of a arc furnace reactor. They are used to improve operating regime of furnace transformers in iron and steel plants [14].

Introduction

Learning Outcomes

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Transformers

- Although transformers are **NOT** an energy conversion device, it is an indispensable component in many energy conversion systems.
- In AC power systems, it makes it possible:
 1. Electric generation at the most economical generator voltage,
 2. Power transfer at the most economical transmission voltage,
 3. Power utilisation at the most suitable voltage.
- The transformer is also widely used in:
 - Impedance matching for maximum power transfer,
 - Isolating one circuit from another,
 - Isolating Direct Current (DC) while maintaining Alternating Current (AC) continuity.

Introduction

Learning Outcomes

Transformers



Figure 11: Delta Transformers specialises in the design and manufacture of dry-type power transformers up to 15,000 kV · A, 34,5 kV class [15].

Introduction

Learning Outcomes

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Drive Technology

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Transformers



- Transformer is one of the simpler devices made of two (2) or more electric circuits coupled by a **common magnetic circuit**.
- Its analysis involves many of the principles essential to the study of electric machinery.

Our study of the transformer will serve as a bridge between magnetic-circuit analysis and the study of electric machinery.

Introduction

Learning Outcomes

Transformers



- The essence of transformer action requires only the existence of **time-varying mutual flux** linking two (2) windings.
- Such action can occur for two (2) windings coupled through air. However, this would be highly inefficient, therefore an iron-core or other ferromagnetic materials are used.

This forces the flux path to be confined to the material.

- Such a transformer is commonly called an **iron-core transformer**.

Most transformers are of this type which will be our focus.

Introduction

Learning Outcomes

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Transformers



- Essentially, a transformer consists of two (2) or more windings coupled by **mutual magnetic flux**.
- If primary is connected to an AC-voltage source, an AC flux will be produced whose amplitude will depend on:
 - The primary voltage (v_{in}),
 - The primary voltage frequency (f_1),
 - Number of turns (N_1),
- The mutual flux will link the secondary, and will **induce** a voltage in it whose value will depend on the number of secondary turns as well as the magnitude of the mutual flux and the frequency.

By changing the number of primary and secondary turns, almost any voltage ratio can be obtained.

Introduction

Learning Outcomes

Transformers



An ideal transformer is a **lossless device** with:

- An input winding (N_1)
- An output winding (N_2)

It has the following properties:

- No iron and copper losses.
- No leakage fluxes.
- A core of infinite permeability and infinite electrical resistivity.
- Flux is confined to the core and winding resistance are negligible.

Ideal Transformer

Deriving the Rules

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Transformers

- The ideal transformer is defined from **Faraday's law**:

Change in magnetic field will create emf ($\mathcal{E}$).

$$v_1 = \mathcal{E}_1 = -\frac{d\lambda}{dt} = -\frac{d(N_1\phi)}{dt} = -N_1\frac{d\phi}{dt}.$$

- The core flux also links the secondary and produces an induced Electromotive Force (EMF) $\mathcal{E}_2$, and an equal secondary terminal voltage v_2 , given by:

$$v_2 = \mathcal{E}_2 = -\frac{d\lambda}{dt} = -\frac{d(N_2\phi)}{dt} = -N_2\frac{d\phi}{dt}.$$

Ideal Transformer

Deriving the Rules

Transformers

- Adding load creates current i_2 and MMF $N_2 i_2$ on the secondary.
- As it is ideal, it has infinite permeability ($\mu \rightarrow \infty$) which means the flux is **unchanged** between primary and secondary.

$$N_1 i_1 - N_2 i_2 = 0$$

- Rearranging it we derive:

$$\frac{i_1}{i_2} = \frac{N_2}{N_1} \quad \blacksquare$$

This means in an ideal transformer $P_{in} = P_{out}$.

Ideal Transformer

Deriving the Rules

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Transformers

- As flux is present in both v_1, v_2 we can derive the following:

$$\frac{v_1}{v_2} = \frac{N_1}{N_2}$$

An ideal transformer transforms voltages in the **direct ratio** of the turns in its windings.

Ideal Transformer

Deriving the Rules

Transformers

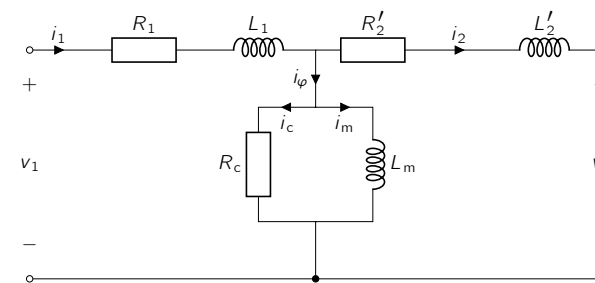


Figure 12: The equivalent circuit for a transformer.

Ideal Transformer

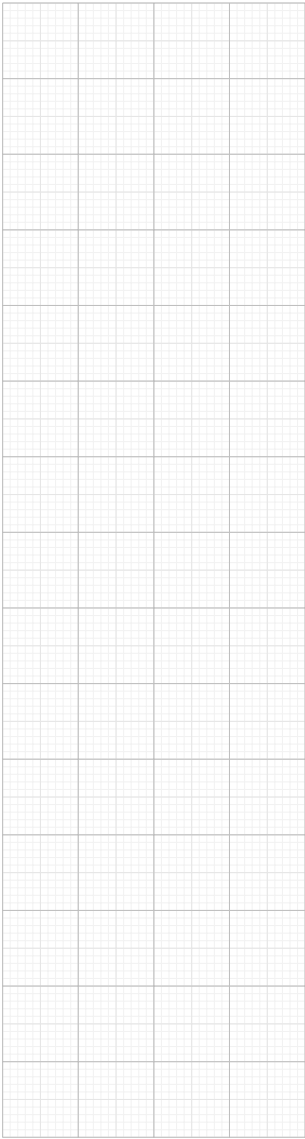
Equivalent Circuit

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Transformers

- The open-circuit test (no-load test), is one of the methods used to determine the no-load impedance in the excitation branch of a transformer.
- The test is done on the **primary side** of the transformer.
- The wattmeter, ammeter and the voltage are connected to their primary winding.
- The nominal rated voltage is supplied to their primary winding.

As the no-load current is very small, the copper losses experienced are negligible.

Parameter Tests

Open Circuit Test

Transformers

Calculation

- Let:

W_0 wattmeter reading,

V_1 voltmeter reading,

I_0 ammeter reading.
- We can state, the iron loss of the transformer (P_C) as:

$$W_0 = P_C = V_1 I_0 \cos \varphi_0$$

The no-load power factor ($\cos \varphi_0$) is then:

$$\cos \varphi_0 = \frac{W_0}{V_1 I_0}$$

Parameter Tests

Open Circuit Test

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Transformers

- The secondary winding of the transformer is **kept open**, and the voltmeter is connected to their terminal.

This voltmeter measures the secondary induced voltage.

- As the secondary of the transformer is open, therefore no-load current flows through the primary winding.
- The value of no-load current is very small as compared to the full rated current.
- The copper loss occurs only on the primary winding of the transformer because the secondary winding is open.
- Wattmeter reading only represents the core and iron losses.
- The core loss of the transformer is the same for all types of loads.

Parameter Tests

Open Circuit Test

Transformers

- The real part of the current $\hat{I}_0$ is:

$$\hat{I}_0 = I_0 \cos \varphi_0, \quad \text{and} \quad \hat{I}_0 = I_0 \sin \varphi_0,$$
- Using these derivations, the core resistance and magnetising inductances are:

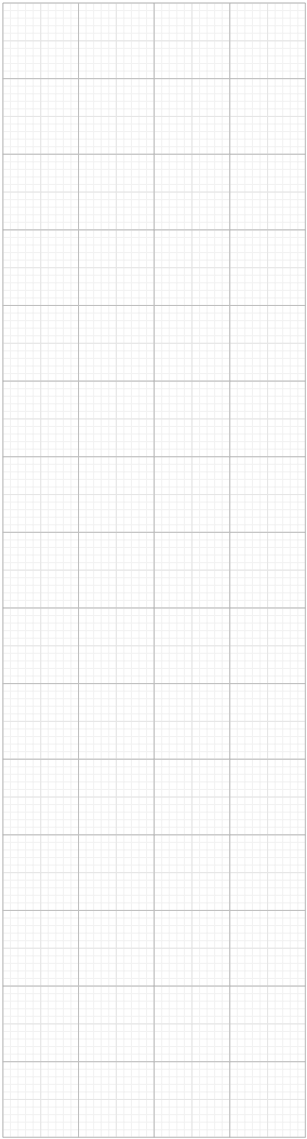
$$R_C = \frac{V_1}{I_m} \quad \text{and} \quad X_m = \frac{V_1}{I_C} \quad \blacksquare$$

Parameter Tests

Open Circuit Test

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Transformers



- The low voltage source is applied across the secondary winding, which is approximately 5 to 10% of the normal rated voltage.
- The flux is set up in the core of the transformer.

The magnitude of the flux is small as compared to the normal flux.

- The iron loss of the transformer depends on the flux.
- This is negligible as the generated flux is very low.
- Wattmeter only determines the copper loss occurred, in their windings.

Parameter Tests

Short Circuit Test

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Transformers



- Wattmeter, voltmeter and ammeter are connected to the high voltage winding.
- Using a variac, the applied voltage is slowly increased until the ammeter gives a reading equal to the rated current of the HV side.
- The full load current is measured by the ammeter connected across their secondary winding.

Parameter Tests

Short Circuit Test

Transformers

- During its operation, a transformer experiences **unavoidable** losses.
- These can be categorised as:
 - Iron (Core) Loss,
 - Copper Loss,
 - Stray Loss,
 - Dielectric Loss.
- These losses appear in the form of heat and causes two (2) problems:
 - Increases transformer temperature,
 - Reduces overall efficiency.

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Transformers	
■ Heat is generated in the transformer while running and is produced by the excitation of the windings and core. ■ If the temperature of the transformer continues to increase rapidly, it results in the degradation of the various parts, specifically the insulation materials, and could lead to the failure of the equipment. ■ Depending on the cooling methods used, transformers can be divided into two (2) types: 1. dry, 2. oil.	
Physical Properties	Cooling
Transformers	
Oil-Type Cooling ■ Uses either: 1. Oil-air cooling, 2. Oil-water cooling. ■ Looking at them in more detail: <i>Oil Natural Air Natural</i> ■ The core and coils are cooled by surrounding in oil. ■ Heat transfer of oil by natural air convection. <i>(Non-Mineral) Oil Natural Air Natural</i> ■ The core and coils are cooled by surrounding in synthetic oil. ■ Heat transfer of oil by natural air convection.	
Physical Properties	Cooling

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Transformers	
Dry-Type Cooling ■ Dry-type transformers, are normally cooled by air. ■ The following two (2) methods adopted in dry-type transformers. Air Natural Cooled by surrounding air via convection. Air Force Forced air circulation using fans and blowers.	
Physical Properties	Cooling

Transformers	
Oil-Type Cooling ■ Uses either: 1. Oil-air cooling, 2. Oil-water cooling. ■ Looking at them in more detail: <i>Oil Natural Air Forced</i> ■ Cooled by surrounding in oil. ■ Forced air circulation using pumps, fans and blowers. <i>Oil Forced Air Forced</i> ■ Forced oil and air circulation using fans and blowers.	
Physical Properties	Cooling

Drive Technology

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- Uses either:

1. Oil-air cooling,
 2. Oil-water cooling.
- Looking at them in more detail:

- Cooled by surrounding in oil.
- Forced water circulation using heat exchanges.

- Forced oil and water circulation using oil-to-water heat exchangers.

Cooling

- A transformer core is composed of **limbs** and **yokes** joined together to form a single structure around which the coils are placed.

The manner in which the respective yoke and limbs join together will depend on the type and design of the core.

Limb Vertical sections which the coils are formed around. The limbs can also be located on the exterior of the outermost coils in the case of some core designs. The limbs on a transformer core can also be referred to as legs.

Yoke The yoke is the horizontal section of the core which joins the limbs together. The yoke and limbs form a pathway for magnetic flux to flow freely.

Core Design and Material

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Figure 13: An example of oil cooled transformer which uses radiators to exchange heat [16].

Cooling

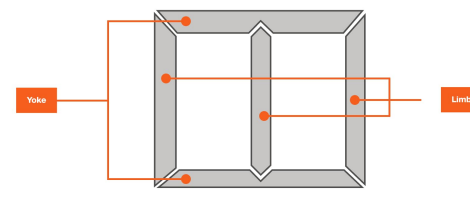


Figure 14: Diagram of the transformer geometry and its parts [17].

Core Design and Material

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Drive Technology

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Transformers



- The earliest transformer cores utilised **solid iron**.
- Methods developed over the years to refine raw iron ore into more permeable materials such as silicon steel, which is used today for transformer core designs due to its **higher permeability**.
- Also, the use of many densely packed laminated sheets reduces issues of circulating currents and overheating caused by solid iron core designs.
- Further increases in core design are made through cold rolling, annealing, and using grain oriented steel.

Physical Properties

Core Design and Material

Transformers



Shell-Type

- The core surrounds the windings.
- Creates a closed pathway surrounding the windings for magnetic flux.
- This design also typically gives less energy loss than core type.
- A shell type design is the classification for most distribution class padmounts and substations with a wrapped 5-legged core.

Core-Type

- A core type design is where the windings surround the core steel.
- In this design, there is no return path (or closed loop) for the magnetic flux around the coils.
- This design typically yields more energy losses, and it requires more copper or aluminium winding material than a shell-type configuration.

Physical Properties

Structure Type

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Transformers



Cold Rolling

- Silicon steel is a softer metal.
- They also have small hysteresis area, low core loss and high μ .
- They are used in making sheets less than 2mm thick.
- Cold rolling silicon steel will increase its strength-making it more durable when assembling the core and coils together.
- Permeability can be increased even further by orienting the grain of the steel in the same direction.

Annealing

- Involves heating core steel to high temperature to remove impurities.
- This process will increase the softness and ductility of the metal by decreasing the internal stress of the material.

Physical Properties

Metallurgical Operations

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Transformers

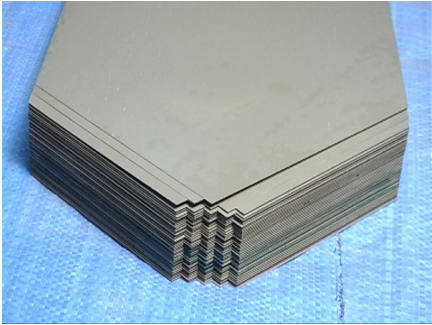


Figure 15: Lamination is done to restrict the eddy current flow.

Physical Properties

Lamination

Transformers

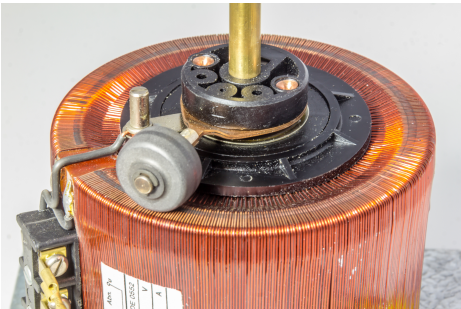


Figure 16: Variable autotransformer 0-220 V, 4 A, 880 VA.

Physical Properties

Transformer Applications

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Transformers

Auto Transformer

- It is a transformer with only one (1) winding.
- portions of the same winding act as both the primary winding and secondary winding sides of the transformer.
- Has higher efficiency than two winding transformer.
 - Less ohmic loss and core loss due to reduction of transformer material.

They are often smaller, lighter, and cheaper than typical dual-winding transformers, but the disadvantage of not providing electrical isolation between primary and secondary circuits.

Physical Properties

Transformer Applications

Transformers

Isolation (Galvanic) Transformer

- Used in transferring power from an AC source to an equipment while isolating the powered device from the power source.
 - Usually for safety reasons or to reduce transients and harmonics.

They provide galvanic isolation where no conductive path between source and load happens.

- This isolation is used to protect against electric shock, to suppress electrical noise in sensitive devices, or to transfer power between two circuits which must not be connected.

Physical Properties

Transformer Applications

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Transformers

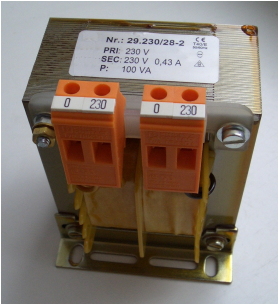


Figure 17: A 230 V isolation transformer.

Physical Properties

Transformer Applications

Transformers

Star-Delta Connection

- Y- Δ connection is commonly used in stepping down from a high voltage to a medium or low voltage due to grounding on the high-voltage side.
- This is due to the neutral connection being available on the high-voltage side for safety reasons.

Delta-Star Connection

- Δ -Y connection is commonly used for stepping up to a high voltage.

Having a neutral connection on the high-voltage side is always a benefit.

Poly-Phase Transformers

Connection Types

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Transformers

- Three single-phase transformers can be connected to form a three-phase transformer bank in any of the four (4) ways:
 - Y- Δ Connection,
 - Δ -Y Connection,
 - Δ - Δ Connection,
 - Y-Y Connection.
- The windings at the left are the **primaries**,
- The ones at the right are the **secondaries**,

Rated V and I at the primary and secondary of the three-phase transformer bank depends upon the connection used but that the rated kVA of the three-phase bank is three times that of the individual single-phase transformers, regardless of the connection.

Poly-Phase Transformers

Connection Types

Transformers

Delta-Delta Connection

- The $\Delta - \Delta$ connection has the advantage where one transformer can be removed for repair or maintenance while the remaining two continue to function as a three-phase bank with the rating reduced to 58 percent of that of the original bank
- This is known as the open-delta, or V, connection.

An open delta transformer is a three phase transformer that only has two primary and secondary windings, with one side of the delta phase diagram "open".

Poly-Phase Transformers

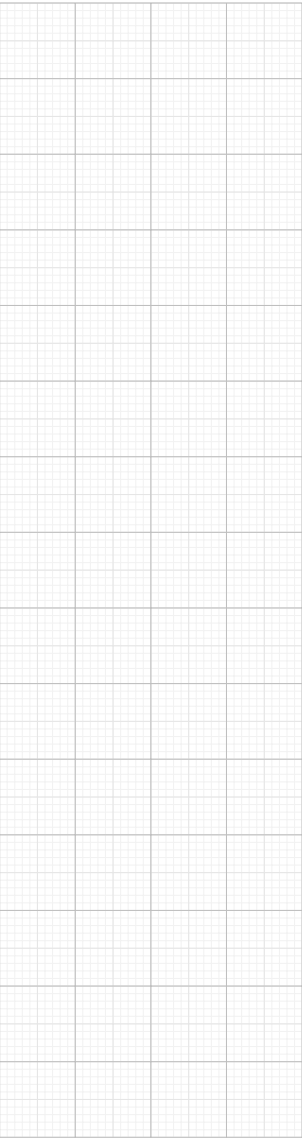
Connection Types

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Transformers

Star-Star Connection

- Seldom used due to significant disadvantages.

Disadvantages

- If the neutral connection is not provided and an unbalanced load is connected, the phase voltages tend to become severely unbalanced.
- Magnetising current varies non-sinusoidal having a third harmonic.
 - In balance, the 3rd harmonic primary winding magnetising currents are equal in magnitude and in phase with each other.
 - These will be directly additive at the neutral point.
- These components will distort the magnetic flux which induces a voltage having a third harmonic component in both windings.
- Third harmonic of induced voltage may be as large as the fundamental.

Poly-Phase Transformers

Connection Types

Electromechanical Energy Conversion

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Transformers

- Instead of three single-phase transformers, a three-phase bank may consist of one three-phase transformer having all six windings on a common multi-legged core and contained in a single tank.
- Advantages of three-phase transformers over connections of three single-phase transformers are that they cost less, weigh less, require less floor space, and have somewhat higher efficiency.

Poly-Phase Transformers

Connection Types

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Learning Outcomes

Forces and Torques in B-Field

- Lorentz Force Law

Energy Balance

W and F in Singly Excited B-Field

Determining Magnetic Force and Energy

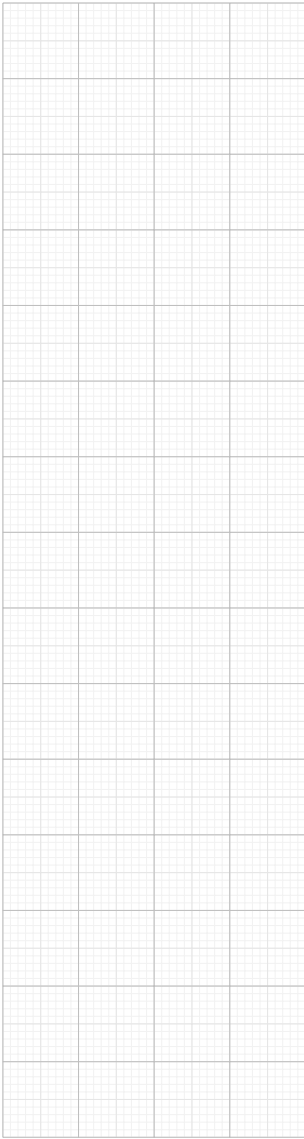
- Energy
- Co-Energy

Multiply Excited Systems

Electromechanical Energy Conversion

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- (L01) Definition of Electromechanical devices,
- (L02) Understanding the conversion process,
- (L03) Describing the energy co-energy relationship,
- (L04) Looking at multiply-excited system..



Learning Outcomes

- The purpose of studying EEC is threefold:
 1. To understand how energy conversion takes place,
 2. To provide techniques for designing and optimising the devices for specific requirements,
 3. To develop models of EEC devices that can be used in analysing their performance as components in engineering systems.

Learning Outcomes

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- The electromechanical energy conversion (EEC) process, takes place through the medium of the electric or magnetic field of the conversion device.
- We can classify these items into three (3) categories:
 1. Devices for measurement and control are frequently referred to as **transducers**.
 - Generally operate under linear input-output conditions and with relatively small signals.
 - i.e., microphones, pickups, sensors, and loudspeakers
 2. Devices encompasses force-producing motions: solenoids, relays, and electromagnets.
 3. Continuous energy-conversion equipment: motors and generators.

Learning Outcomes

- The Lorentz force law states:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

where $\mathbf{F}$ is the force on a particle, q is the charge. $\mathbf{B}$ is the magnetic field, $\mathbf{E}$ is the electric field and $\mathbf{v}$ is the speed in which the particle is moving.

- In a pure E field, the force is simply:

$$\mathbf{F} = q\mathbf{E}$$

- In a B field the force is slightly more complex.

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$$

Forces and Torques in B-Field

Lorentz Force Law

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Electromechanical Energy Conversion

- The principle of conservation of energy states that energy is neither created nor destroyed.
 - It is only a change in **form**.
- For isolated systems with clear boundaries, this allows to keep track of energy:

the net flow of energy into system across its boundary is equal to the sum of the time rate of change of energy stored in the system.

- For the case of electro-mechanical systems we can write:

Energy Balance

Electromechanical Energy Conversion

- Let's rewrite this equation:

Energy Balance

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Electromechanical Energy Conversion

- For the lossless magnetic energy storage system:

$$dW_{\text{elec}} = dW_{\text{mech}} + dW_{\text{fld}} \quad (1)$$

- In time dt , W_{elec} is given as :

$$dW_{\text{elec}} = ei dt$$

where e is the voltage induced in the electric terminals by the changing magnetic stored energy.

- It is through this reaction voltage that the external electric circuit supplies power to the coupling magnetic field and hence to the mechanical output terminals.

Energy Balance

Electromechanical Energy Conversion

- Consider the electromagnetic relay shown schematically

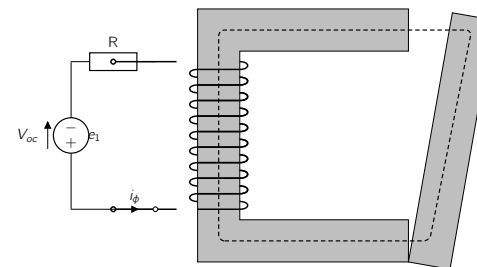


Figure 18: An example of a singly excited magnetic relay.

W and F in Singly Excited B-Field

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Electromechanical Energy Conversion

- The excitation coil resistance is shown as R ,
- The mechanical terminal variables are shown as a force f_{fld} produced by the B -field directed from the relay to the external mechanical system and a displacement x .
- mechanical losses can be included as external elements connected to the mechanical terminal. Similarly, the moving armature is shown as being massless; its mass represents mechanical energy storage and can be included as an external mass connected to the mechanical terminal. As a result, the magnetic core and armature constitute a lossless magnetic-energy-storage system,

W and F in Singly Excited B-Field

Electromechanical Energy Conversion

- The flux linkages λ and current i are considered **linearly dependent** by a geometrically defined **inductance**:

$$\lambda = L(x) i \quad (2)$$

- As magnetic force (f_{fld}) has been defined as acting from the relay upon the external mechanical system and dW_{mech} is defined as the mechanical energy output of the relay, we can write

$$dW_{mech} = f_{fld} dx \quad (3)$$

W and F in Singly Excited B-Field

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Electromechanical Energy Conversion

- As you know, and inductance L is a function of the **geometry** magnetic structure and material.
- EEC devices contain **air-gaps** in their circuits for moving parts.

The reluctance of the air gap is significantly higher than that of the magnetic material which forces most of the energy storage to occur in the air gap.

- This allows us to disregard the **non-linear relations** in the system for practical devices.
- Our second assumption is flux and MMF are directly proportional to the entire magnetic circuit.

W and F in Singly Excited B-Field

Electromechanical Energy Conversion

- Using Eq. (3), and substituting $dW_{elec} = i d\lambda$, we can write:

$$\begin{aligned} dW_{fld} &= dW_{elec} - dW_{mech}, \\ dW_{fld} &= i d\lambda - f_{fld} dx. \end{aligned}$$

- As the magnetic energy storage system is considered **lossless**, it is a **conservative system**.
- The value of W_{fld} uniquely specified by the values of λ and x .
- λ and x are referred to as **state variables** since their values uniquely determine the state of the system.

W and F in Singly Excited B-Field

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Electromechanical Energy Conversion

- We see that W_{fld} is **uniquely** defined by the value of λ and x .

This means the value of W_{fld} is the same regardless of how λ and x are brought to their final values.

- Consider the following paths:

W and F in Singly Excited B-Field

Electromechanical Energy Conversion

- For a lossless magnetic-energy-storage system, the magnetic stored energy W_{fld} is a state function, determined uniquely by the values of the independent state variables λ and x .
- Rewriting Eq. (114) in the following form gives us:

$$dW_{fld}(\lambda, x) = i d\lambda - f_{fld} dx. \quad (4)$$

- For any state function of two (2) independent variables, W_{fld} can be written as:

$$dW_{fld}(\lambda, x) = \left. \frac{dW_{fld}}{d\lambda} \right|_x d\lambda + \left. \frac{dW_{fld}}{dx} \right|_\lambda dx, \quad (5)$$

Determining Magnetic Force and Energy

Energy

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Electromechanical Energy Conversion

- These two separate paths are shown to be integrated to find W_{fld} at the point (λ_0, x_0) .
- The curvy path is the general case and it is difficult to integrate unless both i and f_{fld} are known as a function of λ and x .
- As integration is **path independent** Path 3 gives the same result and is much easier to integrate:

$$W_{fld}(\lambda_0, x_0) = \int_{\text{path 2a}} dW_{fld} + \int_{\text{path 2b}} dW_{fld}$$

W and F in Singly Excited B-Field

Electromechanical Energy Conversion

- As λ and x are **independent** variables, Eq. (4) and Eq. (5) **must** be equal for all values of $d\lambda$ and dx .
- Taking the partial derivative while holding x constant:

$$i = \left. \frac{dW_{fld}(\lambda, x)}{d\lambda} \right|_x \quad (6)$$

- Taking the partial derivative while holding λ constant:

$$f_{fld} = \left. \frac{dW_{fld}(\lambda, x)}{dx} \right|_\lambda$$

Determining Magnetic Force and Energy

Energy

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Electromechanical Energy Conversion

- This is the results we needed.
- Once we know W_{fld} as a function of λ and x , we can solve Eq. (6) as a function for $i(\lambda, x)$

Determining Magnetic Force and Energy

Energy

Electromechanical Energy Conversion

- The coenergy (W'_{fld}) is defined as a function of λ and x :

$$W'_{fld}(\lambda, x) = i\lambda - W_{fld}(\lambda, x)$$

- The desired derivation is carried out by using the differential of $i\lambda$:

$$d(i\lambda) = i d\lambda + \lambda di$$

Determining Magnetic Force and Energy

Co-Energy

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Electromechanical Energy Conversion

- To obtain force, directly as a function of current, we can use co-energy.

Co-energy does not exist in reality. It is a mathematical convenience [18].

- The selection of energy or coenergy as the state function is purely a matter of convenience.
 - They both give the same result, but one or the other may be simpler analytically, depending on the desired result and the characteristics of the system being analysed.

Determining Magnetic Force and Energy

Co-Energy

Electromechanical Energy Conversion

- The analysis of a system with multiple excitation is similar.
- Assume a system having:
 - a mechanical terminal with the values T_{fld} and
 - and two (2) electrical terminals (i.e., $\lambda_1, \lambda_2, i_1, i_2$)

This could be a system where rotor and stator are **both** excited.

- Using the fluxes, the differential energy function $dW_{fld}(\lambda_1, \lambda_2, \theta)$ can be written as:

$$dW_{fld}(\lambda_1, \lambda_2, \theta) = i_1 d\lambda_1 + i_2 d\lambda_2 - T_{fld} d\theta$$

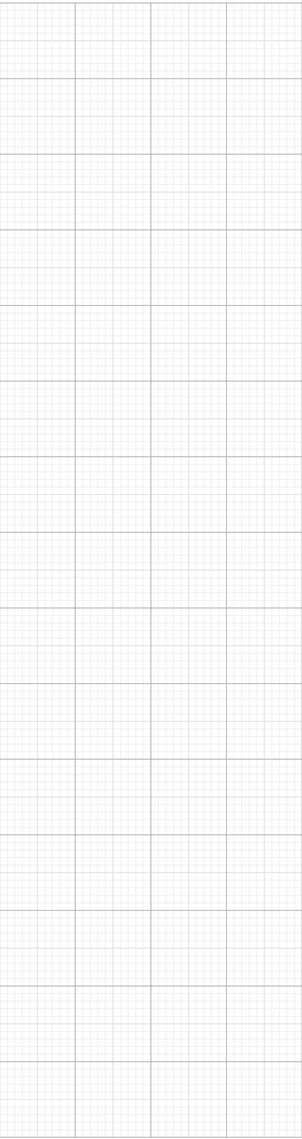
Multiply Excited Systems

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Electromechanical Energy Conversion

- Using our previous analogies:

$$i_1 = \left. \frac{\partial W_{\text{fld}}(\lambda_1, \lambda_2, \theta)}{\partial \lambda_1} \right|_{\lambda_2, \theta}$$
$$i_2 = \left. \frac{\partial W_{\text{fld}}(\lambda_1, \lambda_2, \theta)}{\partial \lambda_2} \right|_{\lambda_1, \theta}$$
$$T_{\text{fld}} = - \left. \frac{\partial W_{\text{fld}}(\lambda_1, \lambda_2, \theta)}{\partial \theta} \right|_{\lambda_1, \lambda_2}$$

The partial derivative with respect to each independent variable must be taken holding the other variables **constant**.

Multiply Excited Systems

Rotating Magnetic Fields

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Electromechanical Energy Conversion

- Using **co-energy** definition, we can re-write our equation as:

$$W'_{\text{fld}}(i_1, i_2, \theta) = \lambda_1 i_1 + \lambda_2 i_2 - W_{\text{fld}}$$

- Taking the derivative of this statement gives us:

$$dW'_{\text{fld}}(i_1, i_2, \theta) = \lambda_1 di_1 + \lambda_2 di_2 - T_{\text{fld}} d\theta$$

- Through significant simplification we arrive at:

$$T_{\text{fld}} = \frac{i_1^2}{2} \frac{dL_{11}(\theta)}{d\theta}$$

Multiply Excited Systems

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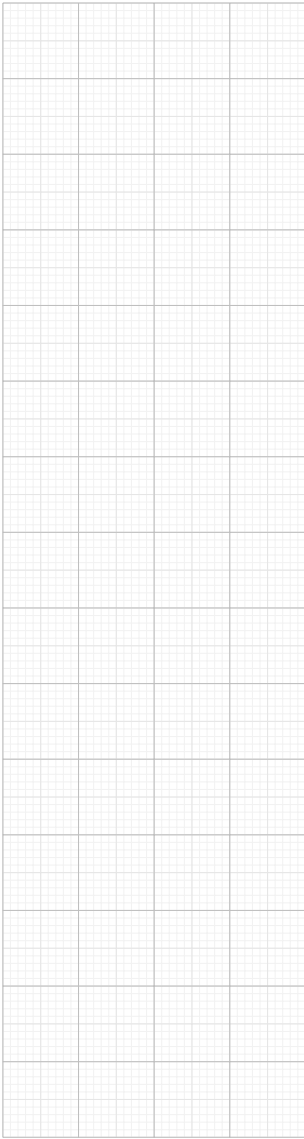
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Rotating Magnetic Fields

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Rotating Magnetic Fields



- Determine the voltages induced by time-varying magnetic fields we use Faraday's law:

$$\mathcal{E} = \frac{d\lambda}{dt},$$

- Conversion occurs when changes in λ results from **mechanical motion**.
- $\mathcal{E}$ are generated in windings or groups of coils by rotating these windings mechanically through a **$\mathbf{B}$** -field, either:
 1. mechanically rotating a magnetic field past the winding,
 2. designing the magnetic circuit so that the reluctance varies with rotation of the rotor (i.e., $\mathcal{R}(\theta)$),

A coil's λ is changed cyclically generating time-varying voltage.

Elementary Concepts

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Rotating Magnetic Fields

- The topic is to introduce and discuss some of the principles underlying the performance of electric machinery.

Following principles are common to both AC and DC machines.

- Various techniques and approximations involved in reducing the machine to simple mathematical models will be discussed.



Introduction

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Rotating Magnetic Fields



Figure 19: The armature of a AC is the stator as the stator has AC flowing during operation [19].

Elementary Concepts

Rotating Magnetic Fields

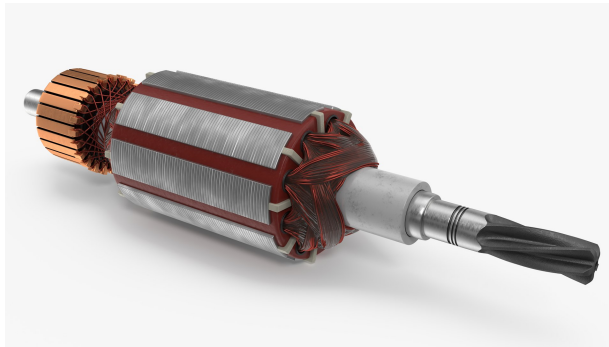


Figure 20: The armature of a DC is the rotor as the rotor has AC flowing during operation [20].

Elementary Concepts

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Rotating Magnetic Fields

- In DC machines, armature winding is found on the rotating member (i.e., rotor).
- The armature winding of a DC machine consists of numerous coils connected together to form a **closed loop**.
- A rotating mechanical contact is used to supply current to the armature winding as the rotor rotates.
 - This is known as a **commutator**.

Elementary Concepts

Rotating Magnetic Fields

- Synchronous Machine (SynM) and DC machines usually include a second winding(s) which carry DC and which are used to produce **main operating flux** in the machine.
- Such a winding is typically referred to as **field winding**.
 - On DC machines, it is found on the stator,
 - On SynM machines, it is found on the rotor.

PMs also produce DC magnetic flux and are used in the place of field windings in some machines.

Elementary Concepts

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- ## Elementary Concepts

AC Machines

Synchronous Machines

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Elementary Concepts

AC Machines

Synchronous Machines

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Synchronous Machines

Synchronous Machines

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Figure 24: A 10,000 HP SynM.

AC Machines

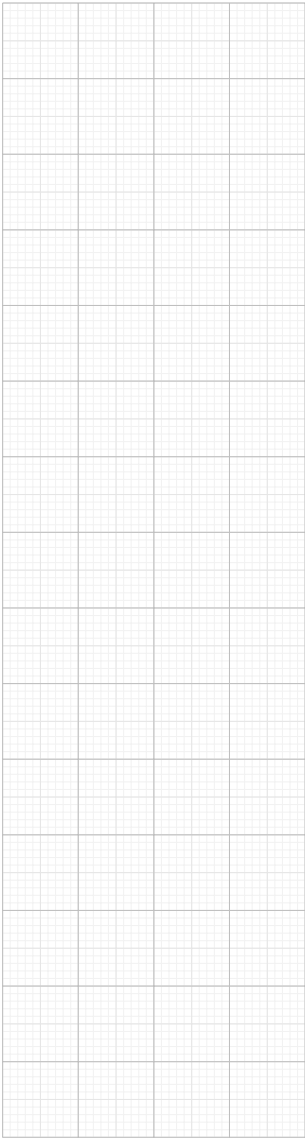
Synchronous Machines

AC Machines

Synchronous Machines

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Rotating Magnetic Fields

- In practice, the air-gap flux-density of practical salient-pole machines can be made to approximate a sinusoidal distribution by properly shaping the pole faces.
- As the rotor rotates, the λ of the armature winding **change with time**.
- Under the assumption of a sinusoidal flux distribution and constant rotor speed, the resulting coil voltage will be sinusoidal in time.
- The coil voltage passes through a complete cycle for each revolution.
- Its frequency (f) in cycles per second (Hz) is the same as the speed of the rotor in revolutions per second: the electric frequency of the generated voltage is synchronized with the mechanical speed, and this is the reason for the designation "synchronous" machine.
- Thus a two-pole synchronous machine must revolve at 3600 revolutions per minute to produce a 60-Hz voltage.

AC MachinesSynchronous Machines

Rotating Magnetic Fields

- Many SynM have more than two poles.
- As a specific example, a schematic form a four-pole single-phase generator.

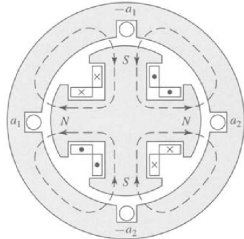


Figure 26:

AC MachinesSynchronous Machines

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Rotating Magnetic Fields

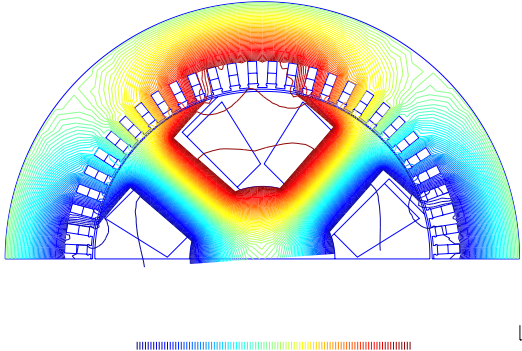


Figure 25: Finite-element solution for the flux distribution around a salient pole.

AC MachinesSynchronous Machines

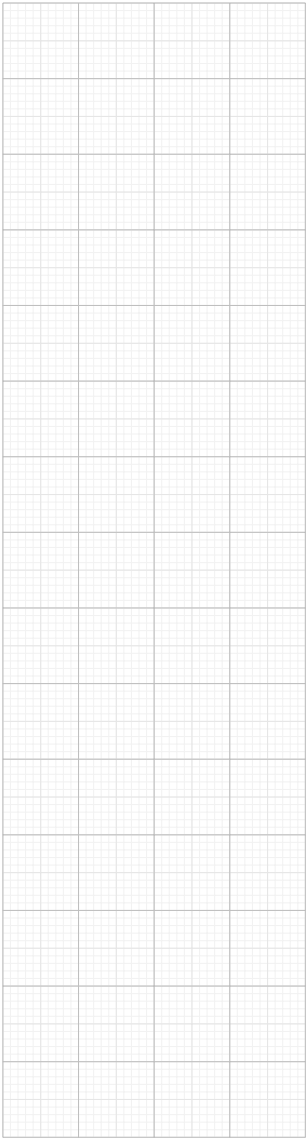
Rotating Magnetic Fields

- The field coils are connected so that the poles are of alternate polarity.
- There are two complete wavelengths, or cycles, in the flux distribution around the periphery.
- The armature winding now consists of two coils connected in series by their end connections.
- The span of each coil is one wavelength of flux.
- The generated voltage now goes through two complete cycles per revolution of the rotor.
- The frequency in hertz will thus be twice the speed in revolutions per second.

AC MachinesSynchronous Machines

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Rotating Magnetic Fields

- When a machine has more than two poles, it is convenient to concentrate on a single pair of poles and to recognize that the electric, magnetic, and mechanical conditions associated with every other pole pair are repetitions of those for the pair under consideration.
- For this reason it is convenient to express angles in electrical degrees or electrical radians rather than in physical units.
- One pair of poles in a multipole machine or one cycle of flux distribution equals 360 electrical degrees.
- Since there are poles/2 complete wavelengths, or cycles, in one complete revolution, it follows, for example, that

$$\theta_e = \left(\frac{p}{2}\right) \theta_a$$

AC Machines

Synchronous Machines

Rotating Magnetic Fields

- Similar to SynM, stator winding of an Induction Machine (IM) is excited with AC.
- Unlike a SynM, rotor field windings are excited with AC.
- In IM, AC is applied directly to the stator windings.
- Rotor currents are produced by induction.
- IM may be regarded as a transformer where electric power is transformed between rotor and stator with a change of frequency and flow of mechanical power.

Most common of all motors, is seldom used as a generator.

- Performance characteristics as a generator are unsatisfactory, although it has been found to be well suited for wind-power applications.

AC Machines

Induction Machines

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Rotating Magnetic Fields

- Armature currents rotates at synchronous speed.
- To produce a steady electromechanical torque, the **B**-fields of the stator and rotor must be **constant in amplitude** and **stationary with respect to each other**.
- In a SynM, the steady-state speed is determined by the number of poles and the frequency of the armature current.

a SynM operated from a constant-frequency AC source will operate at a constant steady-state speed.

AC Machines

Synchronous Machines

Rotating Magnetic Fields

- In IMs, stator windings are **essentially same** as those of a SynM.
- The rotor windings are electrically short-circuited and frequently have no external connections.
- currents are induced by transformer action from the stator winding.
- Here the rotor "windings" are actually solid aluminum bars which are cast into the slots in the rotor and which are shorted together by cast aluminum rings at each end of the rotor.
- This type of rotor construction results in IMs which are relatively inexpensive and highly reliable, factors contributing to their immense popularity and widespread application

AC Machines

Induction Machines

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Rotating Magnetic Fields

- Similar to SynM, the armature flux in the IM leads the rotor and produces torque.
- just as in a SynM, the rotor and stator fluxes rotate in synchronism with each other and that torque is related to the relative displacement between them.
- However, unlike a SynM, the rotor of an IM does not itself rotate synchronously.
- it is the "slipping" of the rotor with respect to the synchronous armature flux that gives rise to the induced rotor currents and hence the torque.
- Induction motors operate at speeds less than the synchronous mechanical speed.

AC Machines

Induction Machines

Rotating Magnetic Fields

- The field winding is on the stator and is excited by direct current.
- The rotor is normally turned at constant speed by a source of mechanical power connected to the shaft.
- The air-gap flux distribution usually approximates a flat-topped wave, rather than the sine wave found in AC machines.
- Rotation of the coil generates a coil voltage which is a time function having the same waveform as the spatial flux-density distribution.

DC Machines

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Rotating Magnetic Fields

- The armature winding of a DC generator is on the rotor with current conducted from it by means of **carbon brushes**.



Figure 27: Brushes are needed for conventional DC Machines to transfer electricity to the rotor.

DC Machines

Rotating Magnetic Fields

- While the purpose is the generation of DC, the voltage induced in an individual armature coil is AC, which must therefore be rectified.
- The output voltage of an ac machine can be rectified using external semiconductor rectifiers.
- This is in contrast to the conventional dc machine in which rectification is produced mechanically by means of a commutator, which is a cylinder formed of copper segments insulated from each other by mica or some other highly insulating material and mounted on, but insulated from, the rotor shaft.
- Stationary carbon brushes held against the commutator surface connect the winding to the external armature terminals.

Commutation is why the armature windings are on rotor.

DC Machines

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Rotating Magnetic Fields

- Most armatures have distributed windings, i.e., windings which are spread over a number of slots around the air-gap periphery.
- The individual coils are interconnected so that the result is a magnetic field having the same number of poles as the field winding.
- The study of the magnetic fields of distributed windings can be approached by examining the magnetic field produced by a winding consisting of a single N-turn coil which spans 180 electrical degrees called a full-pitch coil

MMF Waveform

Rotating Magnetic Fields

- Fourier analysis shows the air-gap MMF produced by a single coil such consists of a fundamental space-harmonic component as well as a series of higher-order harmonic components.
- In AC machine design, serious efforts are made to distribute the coils making up the windings so as to minimise higher-order harmonic components and to produce an air-gap MMF wave which consists predominantly of the space-fundamental sinusoidal component.
- It is thus appropriate here to assume that this has been done and to focus our attention on the fundamental component.

MMF Waveform

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Rotating Magnetic Fields

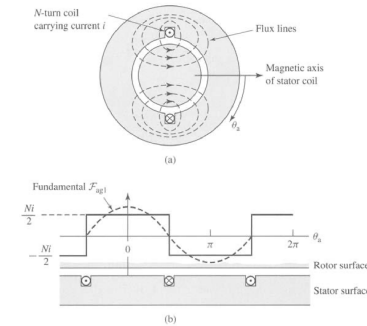


Figure 28:

MMF Waveform

Rotating Magnetic Fields

- The rectangular air-gap MMF wave of the concentrated two-pole, full-pitch coil can be resolved into a Fourier series comprising a fundamental component and a series of odd harmonics.
- The fundamental component ($\mathcal{F}_{ag1}$) is:

$$\mathcal{F}_{ag1} = \frac{4}{\pi} \left(\frac{Ni}{2} \right) \cos \theta_a$$

where θ_a is measured from the **B-axis of the stator coil**.

- It is a sinusoidal space wave of amplitude:

$$(\mathcal{F}_{ag1})_{\text{peak}} = \frac{4}{\pi} \left(\frac{Ni}{2} \right)$$

- with its peak aligned with the magnetic axis of the coil.

MMF Waveform

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Rotating Magnetic Fields

- Electric machinery often contains systems of multiple windings, requiring careful bookkeeping to account for the flux contributions of the various windings.
- Although the details of such analysis are beyond the scope of the lecture, it is useful to discuss these effects in a qualitative fashion and to describe how they affect the basic machine inductances.

Material ConstraintsLeakage Flux

Rotating Magnetic Fields

- Often small, the space-harmonic components of air-gap flux do exist.
- Dc machines they are useful torque-producing fluxes and therefore can be counted as mutual flux between the rotor and stator windings.
- ac machines, however, they may generate time-harmonic voltages or asynchronously rotating flux waves.
- These effects generally cannot be rigorously accounted for in most standard analyses.

Material ConstraintsLeakage Flux

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Rotating Magnetic Fields

Air-Gap Space-Harmonic Flux

- Although single distributed coils create air-gap flux with a significant amount of space-harmonic content, it is possible to distribute these windings so that the space-fundamental component is emphasized while the harmonic effects are greatly reduced.

We can neglect harmonic effects and consider only space-fundamental fluxes in calculating the self and mutual-inductance expressions

Material ConstraintsLeakage Flux

Rotating Magnetic Fields

Slot Leakage Flux

- There are flux components which cross the slot.
- It also forms a component of the leakage inductance of the winding producing it.

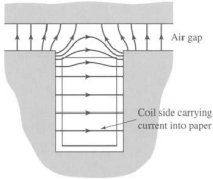


Figure 29: Flux created by a single coil side in a slot.

Material ConstraintsLeakage Flux

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Rotating Magnetic Fields

End-Turn Flux

- The magnetic field distribution created by end turns is extremely complex.
- In general these fluxes do not contribute to useful rotor-to-stator mutual flux, and thus they, too, contribute to leakage inductance.
- Although the leakage inductance is usually difficult to calculate analytically and must be determined by approximate or empirical techniques, it plays an important role in machine performance.

Material Constraints

Leakage Flux

Rotating Magnetic Fields

- Let's add up all these generated magnetic flux densities.

$$\begin{aligned} \mathbf{B}_{\text{net}}(t) &= \mathbf{B}_{RR'} + \mathbf{B}_{YY'} + \mathbf{B}_{BB'}, \\ &= B_{\max} \sin(\omega t) \hat{x} + B_{\max} \sin(\omega t - 120^\circ) \hat{x} + B_{\max} \sin(\omega t - 240^\circ) \hat{x}. \end{aligned}$$

- Each of magnetic field can now be broken to its $\hat{x}$, $\hat{y}$ components.

$$\begin{aligned} \mathbf{B}_{\text{net}}(t) &= B_{\max} \sin(\omega t) \hat{x} - \left[\frac{1}{2} B_{\max} \sin(\omega t - 120^\circ) \right] \hat{x} \\ &\quad + \left[\frac{\sqrt{3}}{2} B_{\max} \sin(\omega t - 240^\circ) \right] \hat{y} + \left[\frac{\sqrt{3}}{2} B_{\max} \sin(\omega t - 240^\circ) \right] \hat{y} \\ &\quad + \left[\frac{\sqrt{3}}{2} B_{\max} \sin(\omega t - 240^\circ) \right] \hat{y}. \end{aligned}$$

Poly-phase RMF

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Rotating Magnetic Fields

- A set of current are applied to the 3-phase stator coils.

$$\begin{aligned} i_{RR'} &= I_{\max} \sin(\omega t), \\ i_{YY'} &= I_{\max} \sin(\omega t - 120^\circ), \\ i_{BB'} &= I_{\max} \sin(\omega t - 240^\circ). \end{aligned}$$

- The magnetic flux densities generated from these currents are;

$$\begin{aligned} \mathbf{B}_{RR'} &= B_{\max} \sin(\omega t) \hat{x}, \\ \mathbf{B}_{YY'} &= B_{\max} \sin(\omega t - 120^\circ) \hat{x}, \\ \mathbf{B}_{BB'} &= B_{\max} \sin(\omega t - 240^\circ) \hat{x}. \end{aligned}$$

Poly-phase RMF

Rotating Magnetic Fields

- Simplification presents us with the following:

$$\mathbf{B}_{\text{net}}(t) = \frac{3}{2} B_m [\sin(\omega t) \hat{x} - \cos(\omega t) \hat{y}] \quad \blacksquare$$

For an AC drive with n phases, the net magnetic vector would be:

$$\mathbf{B}_{\text{net}}(t) = \frac{n}{2} B_m [\sin(\omega t) \hat{x} - \cos(\omega t) \hat{y}] \quad \blacksquare$$

Poly-phase RMF

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Rotating Magnetic Fields



- If the current in any **two of the three coils is swapped**, the direction of the magnetic field's rotation will be **reversed**.
- This means that it is possible to reverse the direction of rotation of an AC motor just by **switching the supply connections on any of two of the three coils**.
- For a 3-phase motor, the rotation of $\mathbf{B}_{\text{net}}(t)$ is:
 - Clockwise:

$$\mathbf{B}_{\text{net}}(t) = \frac{3}{2} B_m [\sin(\omega t) \hat{x} - \cos(\omega t) \hat{y}].$$
 - Counter-clockwise:

$$\mathbf{B}_{\text{net}}(t) = \frac{3}{2} B_m [\sin(\omega t) \hat{x} + \cos(\omega t) \hat{y}].$$

Poly-phase RMF

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